In-App Marketing Impact Measurement

A structured portfolio case study — from business question to measurement framework, analysis, insight, and recommendation

Portfolio disclaimer: This document is a synthetic recreation of a professional analytics project. All data, numbers, SQL, and visualizations are illustrative and created for portfolio demonstration. No confidential, proprietary, customer, or employer-specific information is included.

1. Executive Summary

The purpose of this project was to create a repeatable framework for understanding whether in-app marketing experiences genuinely changed customer behavior. Traditional campaign reporting often focuses on clicks, engagement, or observed conversion. Those metrics can be misleading when users who are already likely to convert are more likely to receive or interact with a campaign.

The framework therefore focuses on incremental conversion lift: the difference in conversion between an exposed treatment cohort and a comparable control cohort. The analysis then breaks the results down by placement and customer segment so marketing teams can determine not only which experiences perform well, but also where and for whom they create the most value.

Illustrative headline finding: In the synthetic dataset, Merchandising Card produced the strongest incremental conversion lift at approximately 5.76 percentage points versus control. These figures are intentionally synthetic and should not be interpreted as actual company performance.

2. Business Problem

Marketing teams use in-app banners, merchandising cards, and pop-ups to influence customer behavior. A campaign may generate strong engagement while contributing little incremental business value, particularly if the exposed audience was already likely to convert.

The central analytical question was therefore:

“Did the in-app marketing experience cause additional customer action that would not otherwise have occurred?”

3. My Role & Ownership

• Framed the business question and converted it into measurable analytical objectives.

• Defined the primary KPI and supporting metrics for evaluating campaign impact.

• Designed the treatment/control measurement approach.

• Built the analytical data structure and reusable SQL transformations.

• Performed user-level, placement-level, and segment-level analysis.

• Used Python for exploratory analysis, calculations, and visualization.

• Translated analytical findings into targeting, placement, and investment recommendations.

• Created a repeatable framework that could be used to evaluate future in-app experiences.

4. Measurement Framework

Primary KPI — Incremental Conversion Lift:
Exposed conversion rate − Control conversion rate

Supporting metrics included engagement rate, conversion rate, revenue, revenue per user, placement performance, and segment-level response. The framework deliberately avoids treating engagement as proof of causality.

5. Analytical Workflow

Step	Purpose
1. Define the question	Translate the business problem into an incremental-impact measurement objective.
2. Create cohorts	Separate exposed users from a control population.
3. Build analytical data	Combine user, campaign, placement, engagement, conversion, and revenue attributes.
4. Measure outcomes	Calculate engagement and conversion rates for each cohort.
5. Calculate lift	Compare treatment performance against control.
6. Segment results	Identify differences by placement and customer segment.
7. Recommend action	Prioritize placements and audiences based on incremental value.

6. Synthetic Dataset

The public portfolio version contains 30,000 synthetic users. Each record includes a user identifier, customer segment, platform, exposure status, in-app placement, conversion outcome, engagement outcome, orders, revenue, and campaign date.

Field	Description
user_id	Synthetic user identifier
segment	New, Returning, High Value, or Reactivated
platform	iOS or Android
exposed	Treatment/control indicator
placement	Banner, Merchandising Card, Pop-up, or Control
engaged	Binary engagement outcome
converted	Binary conversion outcome
orders	Synthetic order count
revenue	Synthetic revenue
campaign_date	Synthetic campaign date

7. Placement-Level Results

Placement	Users	Engagement	Conversion	Lift vs Control (pp)	Revenue / 1k users
Banner	6255	35.17%	26.08%	4.35	13453
Control	14917	23.64%	21.73%	0.0	11357
Merchandising Card	5796	38.34%	27.48%	5.76	14150
Pop-up	3032	32.92%	23.22%	1.49	12603

The synthetic results illustrate why incremental lift is more useful than raw engagement alone. Merchandising Card is the strongest illustrative performer, while Pop-up shows a smaller incremental effect. The control cohort provides the baseline against which each placement is evaluated.

8. Segment-Level Analysis

Placement performance is only part of the decision. A campaign can have a positive overall effect while performing very differently across customer segments. Segment-level analysis therefore helps identify where targeting can create additional efficiency.

Segment	Control Conversion	Exposed Conversion	Lift (pp)
High Value	31.43%	37.82%	6.40
New	16.45%	19.78%	3.33
Reactivated	19.39%	21.54%	2.15
Returning	22.08%	26.59%	4.51

9. Key Insights

• Measure incrementality, not just engagement. A high-engagement placement does not automatically mean high incremental business impact.

• Placement matters. The synthetic analysis shows meaningful differences in incremental conversion across Banner, Merchandising Card, and Pop-up experiences.

• Customer context matters. Segment-level response varies, suggesting that a single experience should not necessarily be shown to every user.

• Targeting and placement should be evaluated together. The strongest opportunity is to match high-impact experiences to segments where they create incremental value.

10. Recommendations

Recommendation	Why
Prioritize placements with positive incremental lift	Allocate more exposure to experiences that demonstrate incremental business value.
Use segment-aware targeting	Concentrate exposure where the treatment creates stronger incremental response.
Avoid click-only optimization	Clicks and engagement can overstate impact when users were already likely to act.
Build repeatable measurement	Use the same framework across future campaigns to create comparable results.
Monitor results continuously	Track lift over time and investigate changes by segment, placement, and platform.

11. Technical Implementation

SQL: Data extraction, transformation, cohort construction, aggregations, and reusable analytical queries.
Python: Exploratory analysis, segment-level calculations, visualization, and analytical automation.
Data layer: The professional project used company data infrastructure; this portfolio version uses synthetic data and SQLite.
Visualization: Python-generated charts; the same outputs can be operationalized in Tableau, Looker, or Power BI.

12. What This Project Demonstrates

This project demonstrates a core approach I use in analytics: start with an ambiguous business problem, define a measurement framework, build a reliable analytical dataset, quantify impact, segment the results, and turn the analysis into a decision.

Business question → Measurement → Data → Analysis → Insight → Recommendation

13. Portfolio Repository

The accompanying GitHub repository contains the synthetic dataset, SQLite database, SQL queries, Python analysis script, charts, README, and this summary document. The repository is intentionally self-contained so a reviewer can reproduce the analysis without access to any proprietary company systems.

14. Confidentiality Note

This portfolio case study does not contain confidential company data, proprietary SQL, internal schemas, customer information, source-system credentials, internal documentation, or employer-specific performance figures. It is designed to demonstrate analytical methodology and storytelling while respecting previous-employer confidentiality.